

Single View Metrology

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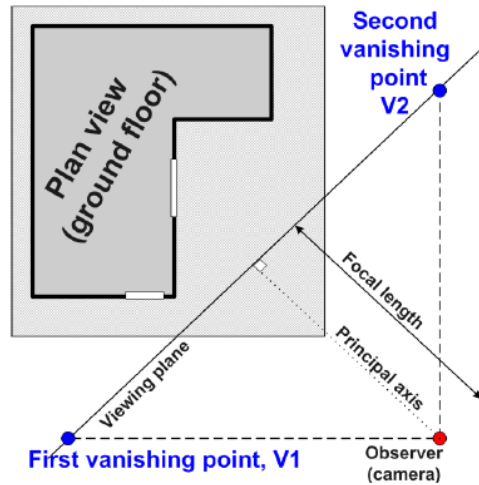
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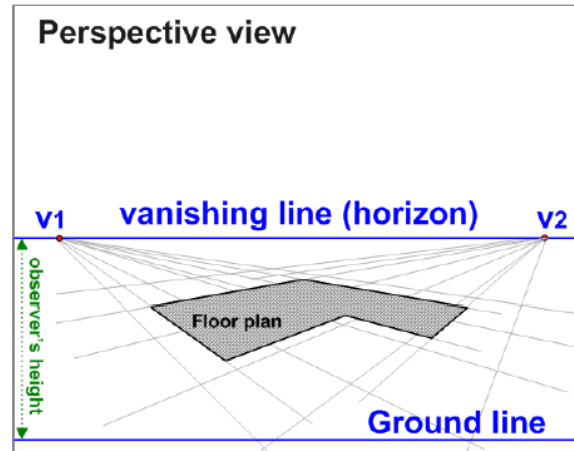
Outline

- Applications
- Projective Geometry
- Height Measurement

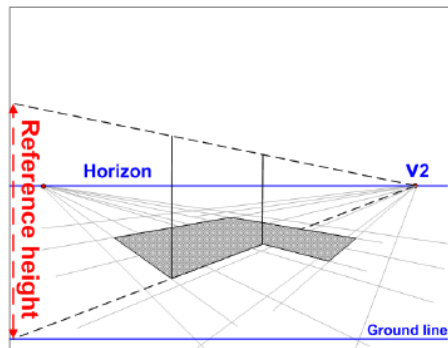
Architectural Drawing



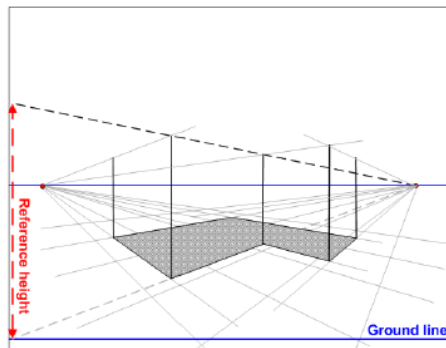
a



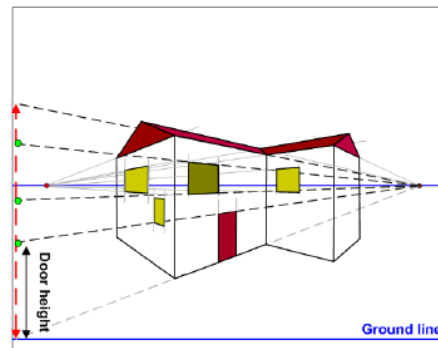
b



c



d



e

- Construct a perspective view of a house from its 3D measurements.
- Conversely, through algebraic modeling provided by projective geometry, 3D measurements can be obtained from flat images.

Forensic Investigation



- Even when the suspect is no longer present in the scene, reference lengths can be measured from fixtures such as tables and windows.

Art History



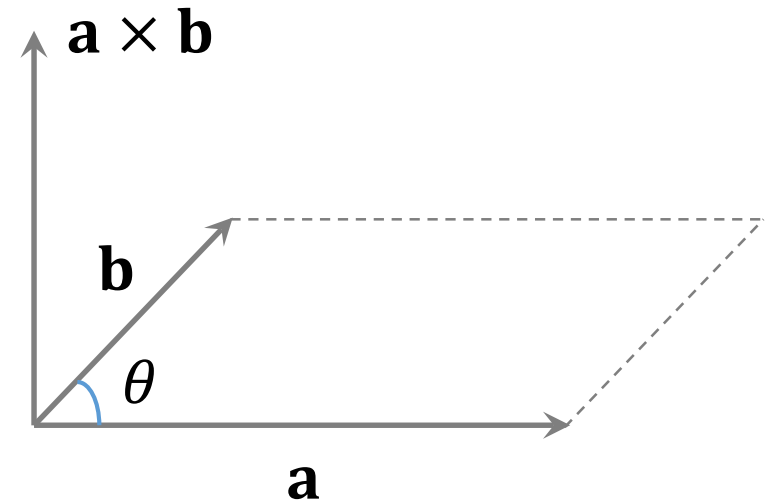
- Single-view metrology can be used to analyze the geometry of paintings.

Outline

- Applications
- **Projective Geometry**
- Height Measurement

Cross Product

- $\mathbf{a} \times \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \sin(\theta) \mathbf{n}$
- $\mathbf{n}$ is a unit vector perpendicular to the plane containing $\mathbf{a}$ and $\mathbf{b}$, in the direction given by [right-hand rule](#).
- Anti-commutative
$$\mathbf{b} \times \mathbf{a} = -(\mathbf{a} \times \mathbf{b})$$
- What is $\mathbf{a} \times \mathbf{b}$ if $\mathbf{a}$ and $\mathbf{b}$ are parallel?



Cross Product Formulas

- $\mathbf{a} \times \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \sin(\theta) \mathbf{n}$

- $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k} = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$

- $\mathbf{a} \times \mathbf{b} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = [\mathbf{a}]_{\times} \mathbf{b}$

- $\mathbf{a}^T (\mathbf{a} \times \mathbf{b}) = \mathbf{b}^T (\mathbf{a} \times \mathbf{b}) = 0$

Homogeneous Representation of Points

- A point $\mathbf{x} = [x, y]^T$ is on the line $\mathbf{l} = [a, b, c]^T \Leftrightarrow ax + by + c = 0$.
- Note that $ax + by + c = [x, y, 1][a, b, c]^T = [x, y, 1] \mathbf{l} = 0$, where the vector $[x, y, 1]$ corresponds to the 2D Cartesian point $[x, y]^T$.
- If $[x, y, 1] \mathbf{l} = 0$ and $k \neq 0$, then $k[x, y, 1] \mathbf{l} = 0$.
- Any vector $[kx, ky, k]^T, k \neq 0$ is a homogeneous representation of $[x, y]^T \in \mathbb{R}^2$.
- An arbitrary homogeneous vector $\mathbf{x} = [x_1, x_2, x_3]^T, x_3 \neq 0$ represents the point $[x_1/x_3, x_2/x_3]^T$ in $\mathbb{R}^2$.
- A homogeneous vector has 2 degrees of freedom, it is represented by 3 elements but has arbitrary scale.

Homogeneous Representation of Lines

- A line in the plane $ax + by + c = 0$ is represented as $[a, b, c]^T$.
- All vectors $[ka, kb, kc]^T = k[a, b, c]^T, k \neq 0$ represent the same line as $[a, b, c]^T$.
- Two vectors $k_1[a, b, c]^T$ and $k_2[a, b, c]^T, k_1 \neq 0, k_2 \neq 0$ are said to be **equivalent**.
- The equivalent class $k[a, b, c]^T, k \neq 0$ is called **homogeneous vectors**.
- The vector $[0,0,0]^T$ does not represent any line.
- The set of homogeneous vectors in $\mathbb{R}^3 - [0,0,0]^T$ forms the **projective space $\mathbb{P}^2$** .
- The notation $-[0,0,0]^T$ means $[0,0,0]^T$ is excluded.

Lines and Points

- The point $\mathbf{x}$ lies on the line $\mathbf{l}$ if and only if $\mathbf{x}^T \mathbf{l} = \mathbf{l}^T \mathbf{x} = 0$.
- The intersection of two lines $\mathbf{l}$ and $\mathbf{l}'$ is the point $\mathbf{x} = \mathbf{l} \times \mathbf{l}'$.
 - Given two lines $\mathbf{l}$ and $\mathbf{l}'$. Define $\mathbf{x} = \mathbf{l} \times \mathbf{l}'$.
 - We have $\mathbf{l}^T (\mathbf{l} \times \mathbf{l}') = \mathbf{l}'^T (\mathbf{l} \times \mathbf{l}') = 0$.
- The line through two points $\mathbf{x}$ and $\mathbf{x}'$ is $\mathbf{l} = \mathbf{x} \times \mathbf{x}'$.
 - Given two points $\mathbf{x}$ and $\mathbf{x}'$. Define $\mathbf{l} = \mathbf{x} \times \mathbf{x}'$.
 - We have $(\mathbf{x} \times \mathbf{x}')^T \mathbf{x} = (\mathbf{x} \times \mathbf{x}')^T \mathbf{x}' = 0$.

Examples of Lines and Points

- Find the intersection of the two lines, $x + y = 2$ and $x = 1$.
- The two lines correspond to $\mathbf{l} = [1, 1, -2]^T$ and $\mathbf{l}' = [1, 0, -1]^T$.
- The intersection point is

$$\mathbf{x} = \mathbf{l} \times \mathbf{l}' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & -2 \\ 1 & 0 & -1 \end{vmatrix} = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}$$

- $\mathbf{x}$ corresponds to the inhomogeneous point $[1, 1]^T$ as required.
- What is the intersection of two parallel lines?

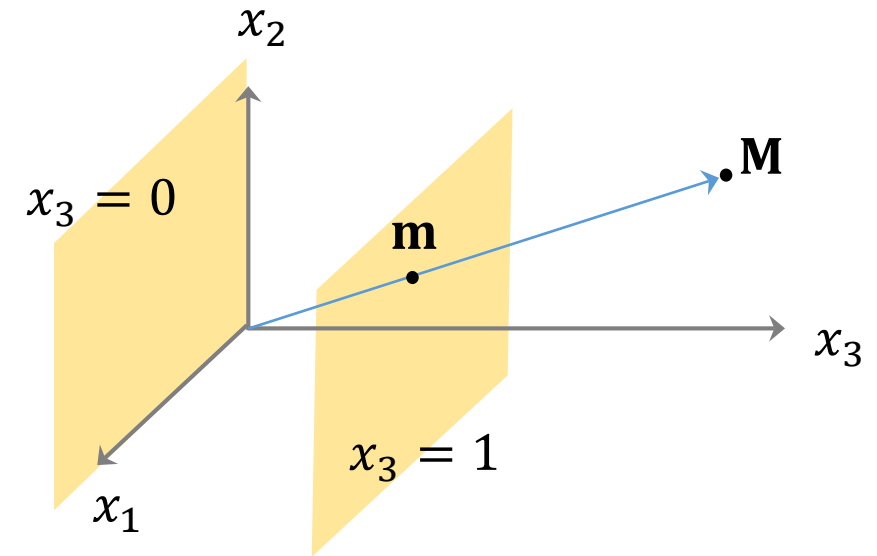
$$\mathbf{x} = \mathbf{l} \times \mathbf{l}' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a & b & c_1 \\ a & b & c_2 \end{vmatrix}$$

Ideal Points & the Line at Infinity

- Homogeneous vectors: $\mathbf{x} = [x_1, x_2, x_3]^T$
- The points with $x_3 = 0$ are known as **ideal points**, or **points at infinity**.
- The **line at infinity** is defined by $\mathbf{l}_\infty = [0, 0, 1]^T$.
- All ideal points lie on the line at infinity, because
$$\mathbf{l}_\infty^T \mathbf{x} = [0, 0, 1][x_1, x_2, 0]^T = 0$$
- The geometry of $\mathbb{P}^n$ is called **projective geometry**.
- In $\mathbb{P}^2$, two distinct lines have exactly one intersection point, independently of if they are parallel or not.
- This is different from the standard Euclidean geometry of $\mathbb{R}^2$.

Interpretation of $\mathbb{P}^2$

- An interpretation of $\mathbb{P}^2$ is as a set of rays in $\mathbb{R}^3$.
- A homogeneous vector $k[x_1, x_2, x_3]^T, k \neq 0$ corresponds to **a ray through the origin**.
- The inhomogeneous representation is given from its intersection with the plane $x_3 = 1$.
- Rays for ideal points lie within the plane $x_3 = 0$ and do not intersect with the plane $x_3 = 1$.



Duality Principle

- The meaning of lines and points are interchangeable in projective geometry.
- The equation $\mathbf{x}^T \mathbf{l} = 0$ may be interpreted as
 - the point $\mathbf{x}$ is on the line $\mathbf{l}$
 - the point $\mathbf{l}$ is on the line $\mathbf{x}$
- The equations $\mathbf{x}^T \mathbf{l} = \mathbf{x}^T \mathbf{l}' = 0$ may be interpreted as
 - the point $\mathbf{x}$ is on the intersection of the lines $\mathbf{l}$ and $\mathbf{l}'$
 - the line $\mathbf{x}$ is through the points $\mathbf{l}$ and $\mathbf{l}'$

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Projection Matrix

Consider the pinhole camera model

$$\tilde{\mathbf{m}}_i = \mathbf{K}[\mathbf{R} \ \mathbf{t}]\tilde{\mathbf{M}}_i$$

Define $\mathbf{P} = \mathbf{K}[\mathbf{R} \ \mathbf{t}]$, then

$$\begin{bmatrix} \lambda_i x_i \\ \lambda_i y_i \\ \lambda_i \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \\ Z_i \\ 1 \end{bmatrix}$$

or

$$\tilde{\mathbf{m}}_i = [\mathbf{p}_1 \ \mathbf{p}_2 \ \mathbf{p}_3 \ \mathbf{p}_4]\tilde{\mathbf{M}}_i$$

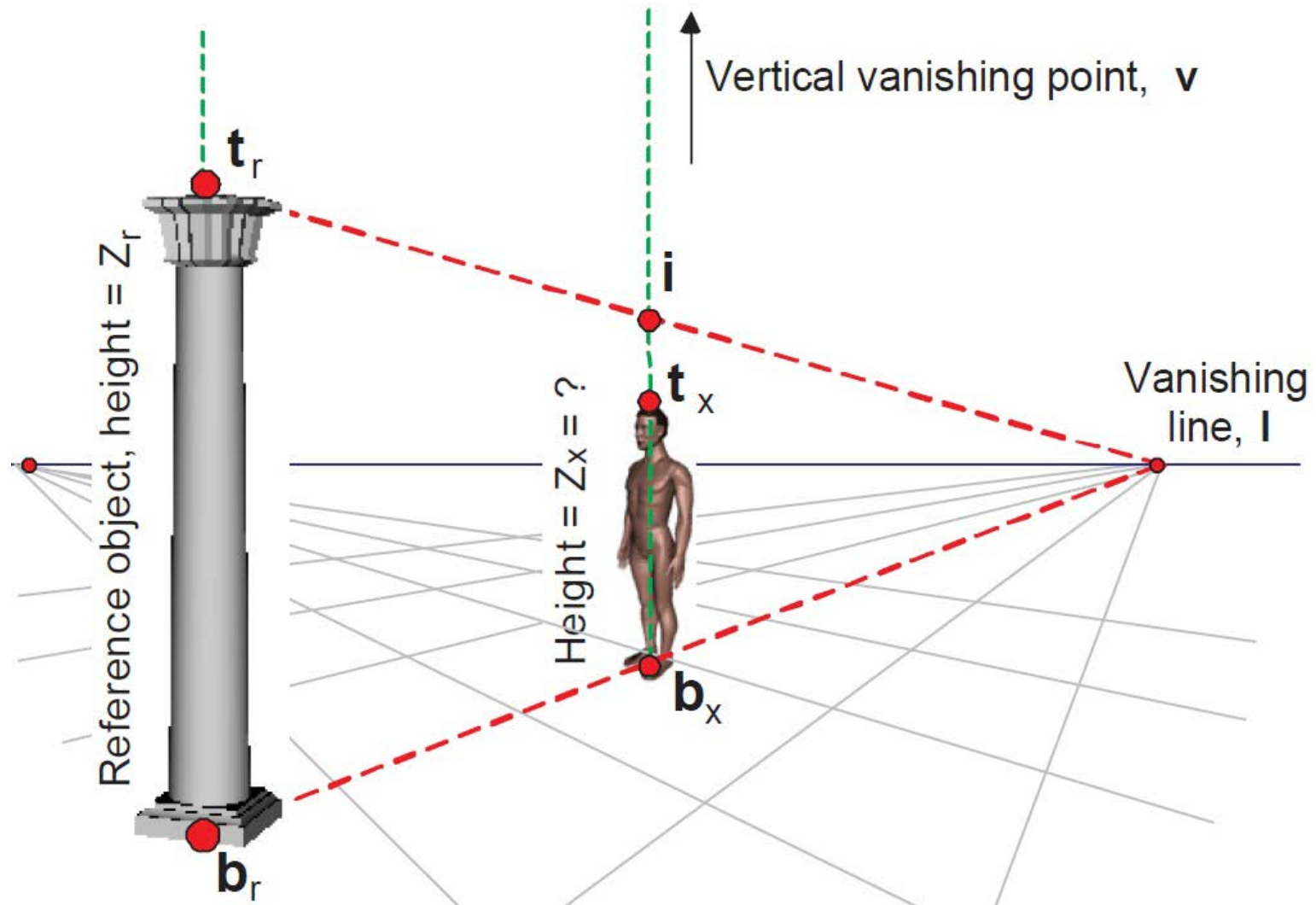
Note that $\mathbf{p}_i$ is a vector in $\mathbb{R}^3$.

Projection Matrix

$$\tilde{\mathbf{m}}_i = [\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3 \quad \mathbf{p}_4] \tilde{\mathbf{M}}_i$$

- Let $\tilde{\mathbf{M}}_0 = [0,0,0,1]^T$, then $\tilde{\mathbf{m}}_0 = \mathbf{p}_4$
 - $\tilde{\mathbf{M}}_0$ is the **world origin**, $\tilde{\mathbf{m}}_0$ is on the vanishing line
 - We can define $\mathbf{p}_4 = \frac{\ell}{\|\ell\|} = \bar{\ell}$
- Let $\tilde{\mathbf{M}}_1 = [1,0,0,0]^T$, then $\tilde{\mathbf{v}}_x = \mathbf{p}_1$, where $\tilde{\mathbf{v}}_x = [\mathbf{v}_x, 1]^T$
 - $\mathbf{v}_x$ is the x-axis **vanishing point**, and λ_x is a scalar
- Let $\tilde{\mathbf{M}}_2 = [0,1,0,0]^T$, then $\tilde{\mathbf{v}}_y = \mathbf{p}_2$, where $\tilde{\mathbf{v}}_y = [\mathbf{v}_y, 1]^T$
- We can define $\mathbf{p}_3 = \alpha \tilde{\mathbf{v}}_z$, where $\tilde{\mathbf{v}}_z = [\mathbf{v}_z, 1]^T$, and α is a scalar.
- Thus, $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \alpha \mathbf{v}, \bar{\ell}]$, where $\mathbf{v} = \tilde{\mathbf{v}}_z$.

Height Measurement



Equation for Height Measurement

$$\alpha Z_{\lambda} = - \frac{\|\mathbf{b}_{\lambda} \times \mathbf{t}_{\lambda}\|}{(\bar{\ell} \cdot \mathbf{b}_{\lambda}) \|\mathbf{v}_z \times \mathbf{t}_{\lambda}\|}$$

- $\mathbf{b}_r$ and $\mathbf{t}_r$ are top and bottom points of the **reference** object ($\lambda = r$)
- $\mathbf{b}_x$ and $\mathbf{t}_x$ are top and bottom points of the **target** object ($\lambda = x$)
- Z_r and Z_x are the heights of the reference and target objects.
- α is the metric factor (a scalar)
- $\bar{\ell} = \ell / \|\ell\|$, where ℓ is the vanishing line
- Note that α is the same for all objects.

Equation for Height Measurement

$$\tilde{\mathbf{m}}_i = \mathbf{P}\tilde{\mathbf{M}}_i = [\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3, \mathbf{p}_4]\tilde{\mathbf{M}}_i = [\mathbf{p}_1, \mathbf{p}_2, \alpha\mathbf{v}, \bar{\ell}]\tilde{\mathbf{M}}_i$$

$$\begin{cases} \mathbf{b}_r = \mathbf{P}[X, Y, 0, 1]^T \\ \mathbf{t}_r = \mathbf{P}[X, Y, Z, 1]^T \end{cases} \Rightarrow \begin{cases} \mathbf{b}_r = \lambda_1(X\mathbf{p}_1 + Y\mathbf{p}_2 + \mathbf{p}_4) \\ \mathbf{t}_r = \lambda_2(X\mathbf{p}_1 + Y\mathbf{p}_2 + Z\mathbf{p}_3 + \mathbf{p}_4) \end{cases}$$

$$\Rightarrow \mathbf{t}_r = \lambda_2 \left(\frac{\mathbf{b}_r}{\lambda_1} + Z\alpha\mathbf{v} \right) \Rightarrow \mathbf{b}_r \times \mathbf{t}_r = -\alpha Z \lambda_1 (\mathbf{v} \times \mathbf{t}_r)$$

Taking the norm of both sides yields

$$\alpha Z = -\frac{\|\mathbf{b}_r \times \mathbf{t}_r\|}{\lambda_1 \|\mathbf{v} \times \mathbf{t}_r\|}$$

In view of $\mathbf{p}_1 \cdot \bar{\ell} = \mathbf{p}_2 \cdot \bar{\ell} = 0$ and $\mathbf{p}_4 \cdot \bar{\ell} = 1$, we have $\mathbf{b}_r \cdot \bar{\ell} = \lambda_1$.

Finally, substitute λ_1 in the denominator with $\mathbf{b}_r \cdot \bar{\ell}$.

Steps of Height Measurement

Given Z_r , the height of the reference object,
We want to compute Z_x , the height of the target object.

1. Label the top and bottom points of the reference object, $\mathbf{b}_r$ and $\mathbf{t}_r$.
2. Label the top and bottom points of the target object, $\mathbf{b}_x$ and $\mathbf{t}_x$.
3. Detect vanishing points $\mathbf{v}_x$, $\mathbf{v}_y$, and $\mathbf{v}_z$ from the image.
4. Compute the metric factor α using the equation for height measurement with the **reference** object.
5. Compute the height Z_x using the equation for height measurement with the **target** object.